1 Elementary Functions

The functions we have encountered in this course so far are called *elementary functions*. These are polynomials, rational functions, power functions (x^a), exponential functions (a^x), logarithmic functions, trigonometric and inverse trigonometric functions, and all functions that can be obtained from these by the five operations or addition, subtraction, multiplication, division, and composition. For instance, the function

$$f(x) = \sqrt{\frac{x^2-1}{x^3+2x-1}} + \ln(\cos(x)) - xe^{\sin(2x)}$$

is an elementary function.

If f is an elementary function, then f' is an elementary function but $\int f(x)dx$ need not be an elementary function. If $\int f(x)dx$ is not an elementary function, then this means that, no matter how hard we try, we will never succeed in evaluating $\int f(x)dx$, at least not in terms of functions that we are familiar with.

Indeed, it has been proved that the following integrals are not elementary functions:

$$\int e^{x^2}dx, \quad \int \frac{e^x}{x}dx, \quad \int \sin(x^2)dx, \quad \int \cos(e^x)dx, \quad \int \sqrt{x^3+1}dx, \quad \int \frac{1}{\ln(x)}dx, \quad \int \frac{\sin(x)}{x}dx.$$

In fact, it turns out that the majority of elementary functions don't have elementary antiderivatives.

Our strategy so far for evaluating the definite integral $\int_a^b f(x)dx$ of a continuous function f has been to find an elementary antiderivative F for f, and then apply the Fundamental Theorem of Calculus to calculate $\int_a^b f(x)dx = F(b) - F(a)$. This strategy is obviously not going to work if $\int f(x)dx$ is not an elementary function, which by the preceding discussion is the case most of the time. As we have also seen, evaluating $\int_a^b f(x)dx$ directly using the definition of the integral is too hard to carry out in practice for all but the simplest functions.

However, note that for many purposes an *approximate* value of $\int_a^b f(x)dx$ is sufficient. In this case, given an integrable function f and an error tolerance ε, we need to produce an estimate E of $\int_a^b f(x)dx$ such that $|E - \int_a^b f(x)dx| < \varepsilon$. The quantity $|E - \int_a^b f(x)dx|$ is called the *absolute error* in our approximation.

We will soon outline several strategies for finding such an approximation. In fact, we have already seen one. Indeed, we began our study of integration by noting that

$$L(f,P) \leq \int_a^b f(x)dx \leq U(f,P)$$

for every partition P of $[a,b]$, so every upper sum and every lower sum is an approximation to $\int_a^b f(x)dx$. However, as we show in the next section, there is actually a lot more flexibility in approximating $\int_a^b f(x)dx$ than just using upper and lower sums.

2 Riemann Sums

Given a bounded function $f : [a,b] \to \mathbb{R}$ and a partition P of $[a,b]$, we now introduce a more general approximating sum to $\int_a^b f(x)dx$ than just the upper and lower sums, called a *Riemann sum*. The additional flexibility of Riemann sums is of great importance in the field of approximate integration, and it pays some modest theoretical dividends as well.

Definition 2.1 (Riemann sum)**.** Let $f : [a,b] \to \mathbb{R}$ be any bounded function and let P be a partition of $[a,b]$.

(a) A choice of a point $x_i^* \in [x_{i-1}, x_i]$ for all $1 \leq i \leq n$ is called a *tagging* of P, which we denote by $\tau = \{x_1^*, \cdots, x_n^*\}$.

(b) A pair (P,τ) is called a *tagged partition*.

(c) Given a bounded function $f : [a,b] \to \mathbb{R}$ and a tagged partition (P,τ) of $[a,b]$, the sum

$$R(f,P,\tau) = \sum_{i=1}^{n} f(x_i^*)(x_i - x_{i-1})$$

is called the *Riemann sum* of f for (P,τ).

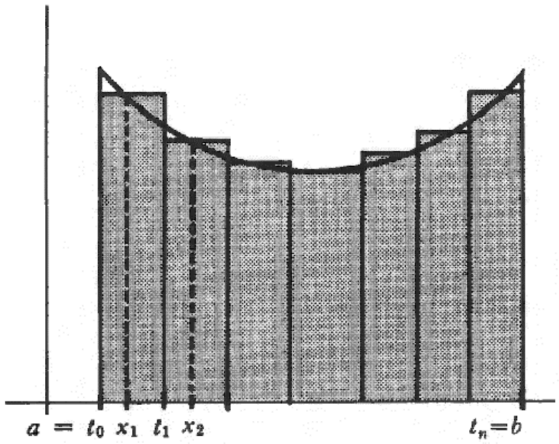


Figure 1: A Riemann sum.

The figure above shows the geometric interpretation of a Riemann sum; it is the total area of n rectangles that lie partly below the graph of f and partly above it. Because of the arbitrary way in which the heights of the rectangles have been picked, we can't say for sure whether a particular Riemann sum is less than or greater than the integral $\int_a^b f(x)dx$. But it does seem that the overlaps shouldn't matter too much; if the bases of all the rectangles are narrow enough, then the Riemann sum ought to be close to the integral. The following theorem makes this precise.

Theorem 2.2. *Suppose f is integrable on $[a,b]$. Then for every $\varepsilon > 0$ there exists $\delta > 0$ such that if P is a partition of $[a,b]$ with $|P| := \max\{x_i - x_{i-1}\} < \delta$ and $\tau = \{x_i^*\}$ any tagging of P, then*

$$\left| R(f,P,\tau) - \int_a^b f(x)dx \right| < \varepsilon.$$

We will prove Theorem 2.2 in a problem of the week.

Corollary 2.3. *Given an integrable function $f : [a,b] \to \mathbb{R}$, choose a partition P of $[a,b]$ consisting of n subintervals of equal width $\Delta x = \frac{b-a}{n}$. Then*

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x = \int_a^b f(x)dx. \tag{2.4}$$

4. Prove Corollary 2.3.

Remark 2.5. A bounded function $f : [a,b] \to \mathbb{R}$ is said to be *Riemann integrable* if there exists a real number I such that for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every partition P with $|P| < \delta$ and all taggings τ of P, $|R(f,P,\tau) - I| < \varepsilon$. In this case, the number I is called the *Riemann integral* of f on $[a,b]$. Theorem 2.2 shows that if f is Darboux integrable, then it is Riemann integrable and $I = \int_a^b f(x)dx$. In fact, it turns out that a function is Riemann integrable if and only if it is Darboux integrable, and that the two integrals coincide. Thus, these two apparently distinct constructions of the definite integral are actually the same, and $\int_a^b f(x)dx$ is usually referred to as the *Riemann integral*. Most calculus textbooks *define* the Riemann integral by Equation (2.4). [1]

[1] In this course, we began with the Darboux integral because is difficult to prove that a continuous function is Riemann integrable directly from the definition; it is much easier to prove that a continuous function is Darboux integrable, and then use Theorem 2.2 to prove that a continuous function is Riemann integrable. Like the Intermediate Value Theorem and Extreme Value Theorem, the theorem that a continuous function is integrable, while central to calculus, is omitted from nearly all calculus textbooks! As you have seen, all of these theorems can be proved using the same simple technique - the Bisection Method.